A kinetic-impact deflection model

antocciphysics

UNIT 3 / DAY 11 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a velocity change and explain the value of lead time.

SUCCESS CRITERIA - I can...

1. State the simplified collision model.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 0.3 kg projectile at 50 m/s sticks to a stationary 2.7 kg cart. Find common speed and the fraction of initial kinetic energy remaining.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × lead time. Toy model: target mass is much larger; straight-line shift comparison. Ejecta can also transfer momentum in a real impact.

Worked reasoning: In a fictional model, a 2 kg projectile at 30 m/s sticks to a stationary 10 kg target. Final speed is 60/12=5 m/s. The system includes both objects.

Your turn: Draw the before-and-after system boundary and momentum arrows.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

After a brief sideways push, a probe has a constant additional sideways velocity. Compare the offset after a short drift and a long drift. Use the model to connect impulse, velocity change and later position. Assume no further sideways force.

Impulse J → Change Δv → Offset Δx

Marzano check - circle: ① ② ③

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Continue down each column.

04 Apply the physics

A hypothetical 600 kg impactor moving at 6000 m/s sticks to a 5 × 10^9 kg target, initially at rest in the target frame. Ignore ejecta. Estimate the target's velocity change and straight-line offset after 10^6 s. Label the system before and after impact. These are model inputs, not mission measurements.

Before | **After**

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An asteroid's crossing window is days 16-20; Earth's is 18-22. Find the overlap and explain why this is not proof of a collision.
Organize the timing evidence before drawing a conclusion.

Object	Arrival time / window	Comparison
A	___	___
B	___	___

Marzano check - circle: 1 2 3

07 Exit check

A model target needs a 10 km shift over 2×10^6 s. Find required Δv. If target mass 10^9 kg, estimate impulse. Organize the known quantities on the before-and-after model.

Before | **After**

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 1000 kg hypothetical target needs a transverse velocity change 0.02 m/s. Estimate required impulse. If delivered uniformly for 0.1 s, find average force and displacement from the velocity change after 10^5 s in a force-free simplified model. Organize the known quantities on the before-and-after model.

Before | **After**

Label objects and signed velocities.

Marzano check - circle: 1 2 3